

JEFF YU

17yuyuncong@gmail.com | 949-527-2256 | Irvine, CA | [linkedin.com/in/jeff-yuncong-yu](https://www.linkedin.com/in/jeff-yuncong-yu) | yuyuncong.com

EDUCATION

University of Southern California – *M.S. in Computer Science*

Expected: Dec 2026

University of California, San Diego – *B.S. in Math & Applied Science*

Jun 2021

TECHNICAL SKILLS

- Languages: Python, Java, C++, C, SQL, MATLAB, HTML/CSS
- Testing & Automation: Playwright, pytest, unittest, Selenium, UI Automation, End-to-End Testing
- AI / ML: Deep Learning Workflows, Computer Vision, NLP, Dataset Benchmark, Model Evaluation, LoRA & SFT
- Technologies: Git, Linux, CI/CD, REST APIs, ArcGIS APIs, Test Framework Design, Object-Oriented Programming

EXPERIENCE

ESRI (Environmental Systems Research Institute)

Redlands, CA

Product Engineer Intern

May 2026 – Aug 2026

- Architected a scalable, maintainable Playwright + pytest automation framework for ArcGIS Enterprise using modular page objects, reusable components, fixtures, and shared utilities to streamline certification testing for AI features.
- Automated end-to-end AI workflows including project creation, training data preparation, model training, inference, and model publishing using Playwright, Python, and ArcGIS APIs to validate both UI behavior and backend state.

OfSpectrum

Los Angeles, CA

Machine Learning Intern

Jun 2025 – Feb 2026

- Created dataset SynSFX, a 178-hour, 43K+ clips benchmark across 7 text-to-audio generators, fine-tuning AASIST to cut EER from 60.84% zero-shot to 2.36–3.23% while uncovering weak generalization to deepfakes from new text-to-audio models.
- Built a scalable Python data generation pipeline using multiprocessing, GPU acceleration, and external APIs to process 250K+ audio artifacts with 30% higher throughput.

Siemens Digital Industries Software

Costa Mesa, CA

Application Engineer

Jul 2021 – May 2023

- Led quality engineering for the NX Architecture Tool Window project, validating redesigned window docking controls, designing regression test strategies, and collaborating with developers to resolve release-blocking defects.
- Developed 60+ automated and interactive test cases across architecture, CAD, and virtual reality workflows using Python and internal automation frameworks, expanding automation coverage and supporting four major software release cycles through CI/CD-integrated regression testing.
- Guided 3 interns through NX automation testing workflows, including test case design, execution, validation, and reporting.

PROJECTS

Geospatial AI Automation Framework | *Python, Playwright, pytest, ArcGIS Enterprise, ArcGIS APIs, REST APIs*

- Engineered reusable infrastructure including Portal API wrappers, GIS map interaction libraries, datastore management utilities, and synchronization helpers, significantly reducing effort required to automate new workflows.
- Built network-aware validation by capturing ArcGIS REST (PBF/JSON) traffic and verifying inference outputs against backend feature and imagery services, providing deterministic validation of geospatial AI workflows.

Low-Rank Adaptation for Mathematical Reasoning | *PyTorch, Hugging Face Transformers, CUDA, QWEN 2.5B*

- Investigated cost-efficient mathematical reasoning fine-tuning on Qwen 2.5B by implementing LoRA adaptation. Achieved 15% higher accuracy than full supervised fine-tuning while updating only ~10% of model weights.
- Implemented a training workflow with dataset preprocessing, tokenizer configuration, prompt formatting, checkpoint management, and evaluation scripts for repeatable fine-tuning experiments.